♂ : Staminate

♀ : Pistillate

⚥ : Bisexual flower

K : Calyx, $\mathbf{K_5}$ five sepals, **aposepalous**, $\mathbf{K_{(5)}}$ five sepals **synsepalous**.

C : Corolla, C_5 five petals, **apopetalous**, $C_{(5)}$ five petals **sympetalous** $\mathbf{C_{(2+3)}}$ corolla bilabiate with upper lib two lobes.

A : Androecium $\mathbf{A_3}$ three stamens free, A_2+_2, Stamens 4, two whorls (**didynamous**) each whorl two stamens (free)

$A_{(9)+1}$ – stamens ten, two bundles **(diadelphous)** 9 stamens unite to form one bundle,1 stamen form another bundle.

$\overset{\frown}{C_5}A_5$—**Epipetalous** represented by an arc.

A^0 :**Staminode**(sterile stamen)

G. Gynoecium or pistil – $\mathbf{G_2}$ – Carpels two, free **(apocarpous)**

$G_{(3)}$ – Carpels three, united **(syncarpous)**

G_0 – pistillode (sterile carpel)

G – superior ovary

G inferior ovary

G– – semi-inferior ovary

∞ – Indefinite number of units

4.7 Fruits

We know about several kinds of fruits, but by botanical study we will be surprised to know the types of fruits and how they are produced by plants. Fruits are the products of pollination and fertilization, usually containing seeds inside. In common person's perspective a fruit may be defined as an edible product of the entire gynoecium and any floral part which is sweet, juicy or fleshy, coloured, aromatic and enclosing seeds. However the fruit is a fertilized and ripened ovary. The branch of horticulture that deals with the study of fruits and their cultivation is called **pomology**.

4.7.1 Structure of Fruit

Fruit has a fruit wall. It is otherwise called **pericarp**. It is differentiated into outer **epicarp**, middle **mesocarp** and inner **endocarp**. The inner part of the fruit is occupied by the seed.

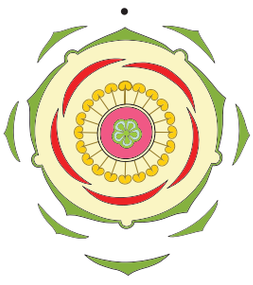


Figure 4.26: (a) *Hibiscus rosa-sinensis*

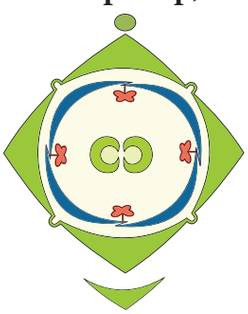


Figure 4.26: (b) *Ixora coccinea*

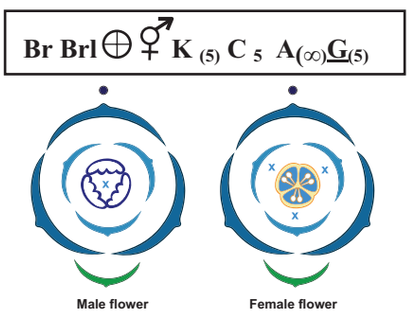


Figure 4.26: (c) *Phyllanthus amarus*

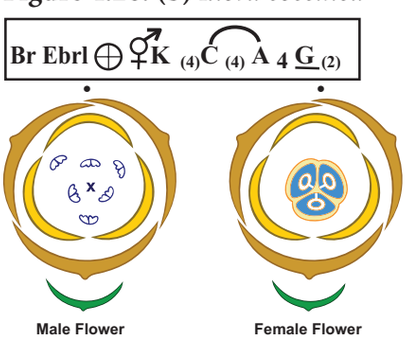
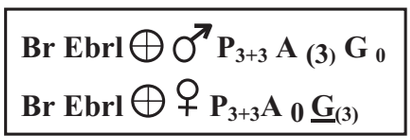
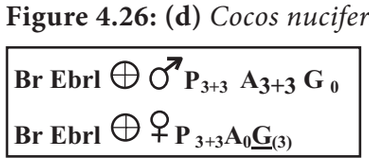


Figure 4.26: (d) *Cocos nucifera*





4.7.2 Types of Fruit

Fruits are classified into three types:

Simple Fruits

The fruits are derived from a single ovary of a flower Example: Mango, Tomato. Simple fruits are classified based on the nature of pericarp as follows.

A. Fleshy Fruit

The fruits are derived from single pistil where the pericarp is fleshy, succulent and differentiated into

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